



Lecture 2

Fourier Series of Functions with Arbitrary Periods

任意周期の関数のフーリエ級数

Outline of Lecture 2

2.1 Fourier Series: Examples

フーリエ級数：例

2.2 Fourier Series of Functions with Arbitrary Periods

任意周期の関数のフーリエ級数

2.1 Fourier Series: Examples

フーリエ級数：例

2.1 Fourier Series: Examples

Arithmetic Average (算術平均)

If we have a data set consisting of the values $a_1, a_2, \dots, a_n$, then the **arithmetic average** A is defined by the formula:

$$A = \frac{1}{n} \sum_{i=1}^n a_i = \frac{a_1 + a_2 + \dots + a_n}{n}$$

https://en.wikipedia.org/wiki/Arithmetic_mean

Recall that f is **piecewise smooth** if f and f' are **piecewise continuous**.

If f is **piecewise continuous**, the **average** (or **arithmetic average**) of f at c is

$$\frac{f(c-) + f(c+)}{2}$$

where $f(c-) = \lim_{x \rightarrow c^-} f(x)$ and $f(c+) = \lim_{x \rightarrow c^+} f(x)$.

If f is **continuous** at c , then $f(c+) = f(c-) = f(c)$ and so the **average** of f at c is $f(c)$.

Average will be of interest **only at points of discontinuity** (不連続).

Example 2.1 The function in Figure 2.1 has a **discontinuity** at $x = 1$. Its **average** is $\frac{1 + \frac{1}{2}}{2} = \frac{3}{4}$.

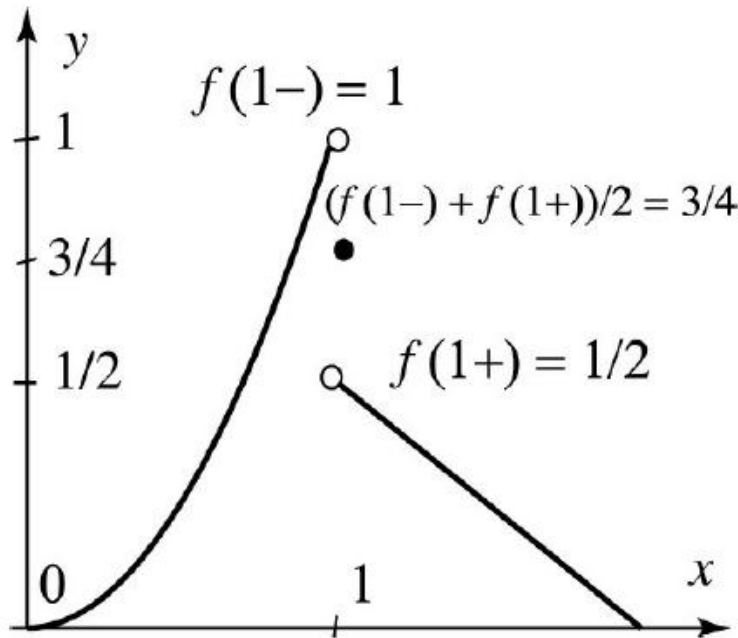


Figure 2.1 Average of $f(x)$ at $x = 1$.

2.1 Fourier Series: Examples

Theorem (定理) 2.1: Fourier Series Representation

Suppose that f is a 2π -periodic piecewise smooth function. Then for all x we have

$$\frac{f(x+) + f(x-)}{2} = a_0 + \sum_{n=1}^{\infty} (a_n \cos nx + b_n \sin nx) \quad (2.1)$$

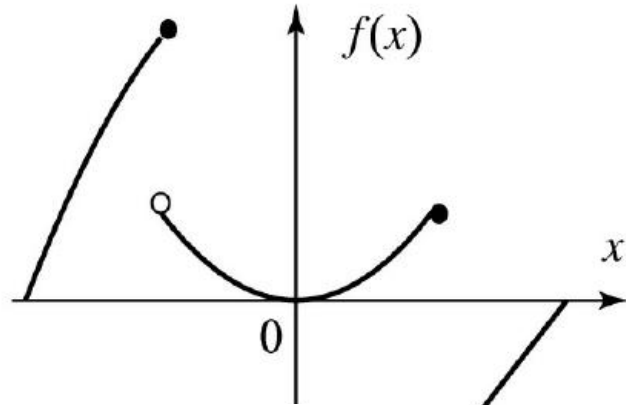
where the Fourier coefficients a_0, a_n, b_n are given by (1.9), (1.10) and (1.11).

In particular, if f is piecewise smooth and continuous at x , then

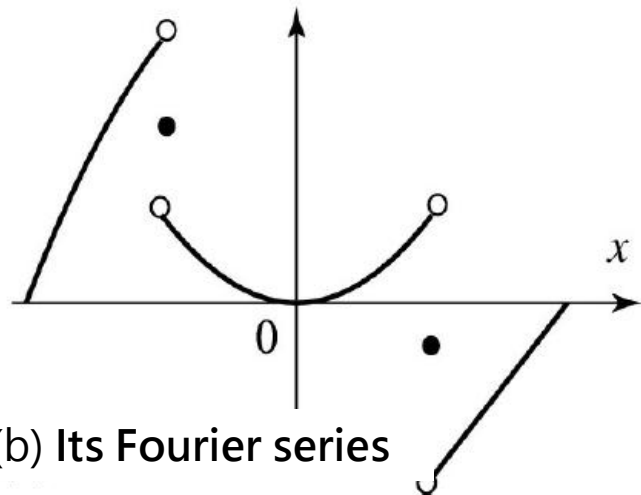
$$f(x) = a_0 + \sum_{n=1}^{\infty} (a_n \cos nx + b_n \sin nx) \quad (2.2)$$

Find the proof in Section 2.8 of the textbook

2.1 Fourier Series: Examples



(a) A piecewise smooth function



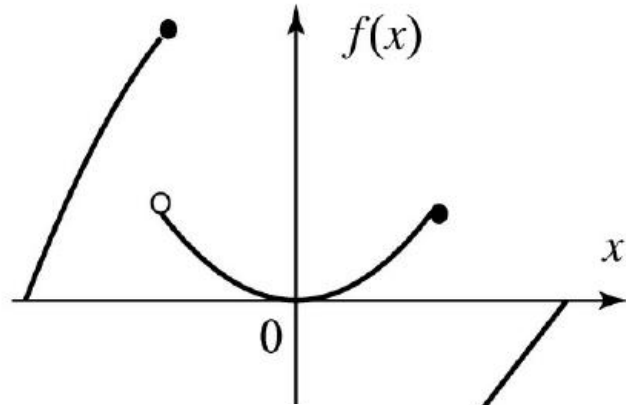
(b) Its Fourier series

Figure 2.2

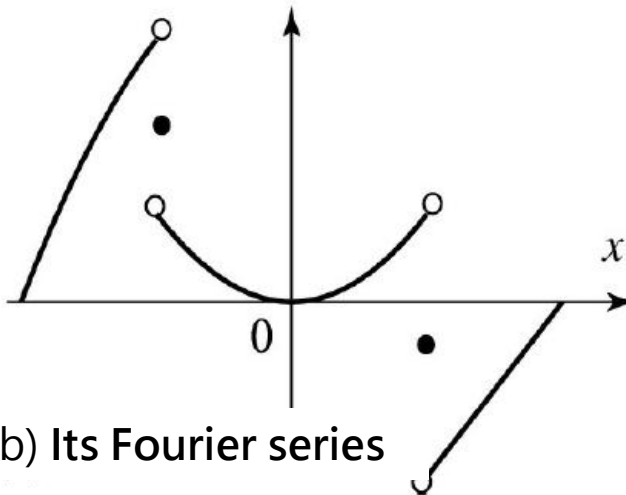
1. At a point of continuity of a piecewise smooth function the Fourier series converges (収束する) to the value of the function.

2. At a point of discontinuity, the Fourier series does its best to converge, and having no reason to favor one side over the other, it converges (収束する) to the average of the left and right limits (see Figure 2.2).

2.1 Fourier Series: Examples



(a) A piecewise smooth function



(b) Its Fourier series

Figure 2.2

For this reason, we may **define** (or redefine) f **at isolated points (孤立点)** without affecting its **Fourier series**.

If we redefine the function at points of discontinuity to be $\frac{f(c-) + f(c+)}{2}$ we then have the equality

$$f(x) = a_0 + \sum_{n=1}^{\infty} (a_n \cos nx + b_n \sin nx)$$

holding at all x .

2.1 Fourier Series: Examples

Even Function (偶関数) and Odd Function (奇関数)

A function f is **even** if $f(-x) = f(x)$ for all x .

A function f is **odd** if $f(-x) = -f(x)$ for all x .

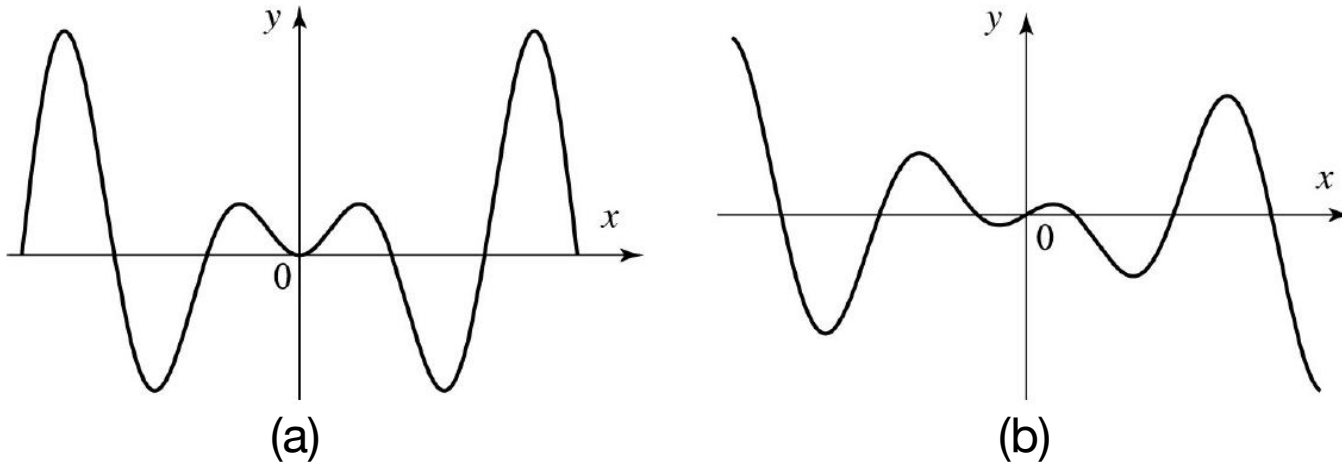


Figure 2.3

(a) **Even** function: The graph is **symmetric** (対称の) with respect to **the y-axis**.

(b) **Odd** function: The graph is **symmetric** with respect to **the origin** (原点).

- If f is even (Figure 2.3(a)), then

$$\int_{-p}^p f(x) dx = 2 \int_0^p f(x) dx \quad (2.3)$$

- If f is odd (Figure 2.3(b)), then

$$\int_{-p}^p f(x) dx = 0 \quad (2.4)$$

2.1 Fourier Series: Examples

Even Function (偶関数) and Odd Function (奇関数)

The following **useful properties concerning the products (積) of these functions** are easily verified.

$$(\text{Even}) \cdot (\text{Even}) = \text{Even}$$

$$(\text{Odd}) \cdot (\text{Odd}) = \text{Even}$$

$$(\text{Even}) \cdot (\text{Odd}) = \text{Odd}$$

- If we have two functions: $f(x) = x^2$ is even, and $g(x) = x^3$ is odd.

$$f(x) \cdot f(x) = x^4 \text{ is even.}$$

$$g(x) \cdot g(x) = x^6 \text{ is even.}$$

$$f(x) \cdot g(x) = x^5 \text{ is odd.}$$

2.1 Fourier Series: Examples

Example 2.2 Triangular wave (三角波)

The 2π -periodic triangular wave is given on the interval $[-\pi, \pi]$ by

$$g(x) = \begin{cases} \pi + x & \text{if } -\pi \leq x \leq 0 \\ \pi - x & \text{if } 0 \leq x \leq \pi \end{cases}$$

- (a) Find its Fourier series.
- (b) Plot some partial sums and the Fourier series.

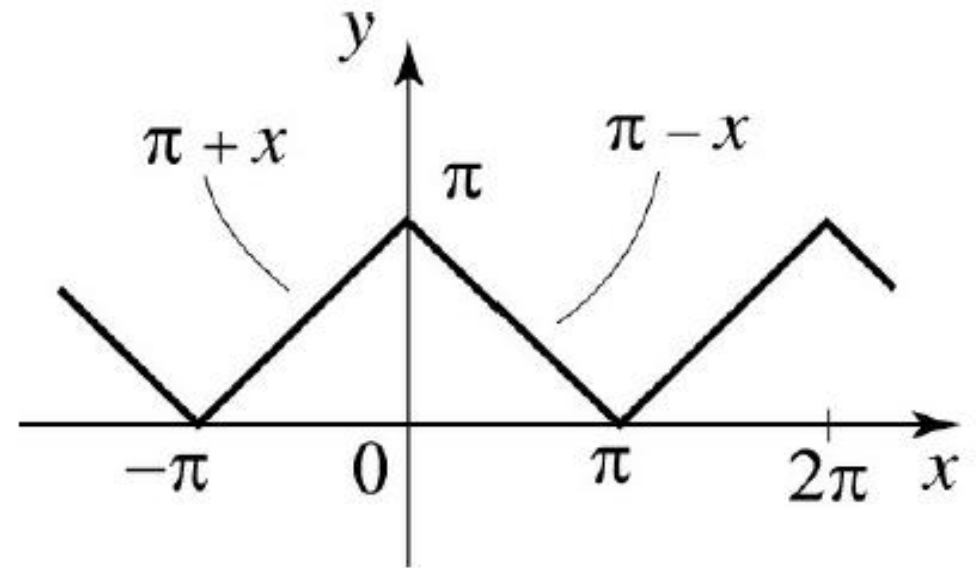


Figure 2.4 Triangular wave

[https://en.wikipedia.org/wiki/Series_\(mathematics\)](https://en.wikipedia.org/wiki/Series_(mathematics))

2.1 Fourier Series: Examples

Solution:

From the figure, we see that $g(x)$ is piecewise smooth and continuous for all x

($g(x)$ is not smooth at $x=k\pi$,

$g'(x)$ does not exist at $x=k\pi$.)

We expect the Fourier series to converge to $g(x)$ for all x , as the second part of Theorem 2.1.

By using (1.9), we have

$$\begin{aligned} \checkmark \quad a_0 &= \frac{1}{2\pi} \int_{-\pi}^{\pi} g(x) dx \\ &= \frac{1}{2\pi} \left[\int_{-\pi}^0 (\pi+x) dx + \int_0^{\pi} (\pi-x) dx \right] \\ &= \frac{1}{2\pi} \left[\left(\pi x + \frac{1}{2} x^2 \right) \Big|_{-\pi}^0 + \left(\pi x - \frac{1}{2} x^2 \right) \Big|_0^{\pi} \right] \\ &= \frac{1}{2\pi} \left[0 - (-\pi^2 + \frac{\pi^2}{2}) + (\pi^2 - \frac{1}{2} \pi^2) \right] \\ &= \frac{1}{2\pi} \pi^2 \\ &= \frac{\pi}{2} \end{aligned}$$

By using (1.10), we have

$$\begin{aligned} \checkmark \quad a_n &= \frac{1}{\pi} \int_{-\pi}^{\pi} g(x) \cos nx dx \\ &= \frac{1}{\pi} \int_{-\pi}^0 (\pi+x) \cos nx dx + \frac{1}{\pi} \int_0^{\pi} (\pi-x) \cos nx dx \\ &\quad \left(\begin{array}{l} \text{change } x \text{ to } -x \\ x = -x \end{array} \right) \\ &= \frac{1}{\pi} \int_0^{\pi} (\pi-x) \cos(-nx) d(-x) = \frac{1}{\pi} \int_0^{\pi} (\pi-x) \cos nx dx \\ &= 2 \cdot \frac{1}{\pi} \int_0^{\pi} (\pi-x) \cos nx dx \\ &= \frac{2}{\pi} \left(\int_0^{\pi} \pi \cos nx dx - \int_0^{\pi} x \cos nx dx \right) \\ &= \frac{2}{\pi} \left(\frac{\pi}{n} \sin nx \Big|_0^{\pi} - \int_0^{\pi} x \cos nx dx \right) \rightarrow \int_0^{\pi} x \cos nx dx \quad \checkmark \\ &\quad \begin{array}{l} u = x \quad dv = \cos nx dx \\ v = \int \cos nx dx = \frac{1}{n} \sin nx \end{array} \\ &= \frac{2}{\pi} \left[0 - \left(\frac{1}{n^2} \cos n\pi - \frac{1}{n^2} \right) \right] \\ &= \frac{2}{\pi} \left[\frac{1}{n^2} - \frac{1}{n^2} \cos n\pi \right] \\ &\quad \int_0^{\pi} u dv = [u \cdot v]_0^{\pi} - \int_0^{\pi} v du \\ &\quad = x \cdot \frac{1}{n} \sin nx \Big|_0^{\pi} - \int_0^{\pi} \frac{1}{n} \sin nx dx \\ &\quad = x \cdot \frac{1}{n} \sin nx \Big|_0^{\pi} + \frac{1}{n^2} \cos nx \Big|_0^{\pi} \\ &\quad = 0 + \frac{1}{n^2} \cos n\pi - \frac{1}{n^2} \end{aligned}$$

Since $\cos n\pi = (-1)^n$, we see that $a_n = 0$ if n is even, and $a_n = \frac{4}{\pi n^2}$ if n is odd.

2.1 Fourier Series: Examples

Solution (cont.)

Finally, by using (1.11), we have

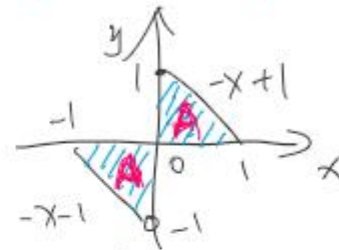
$$\checkmark \quad b_n = \frac{1}{\pi} \int_{-\pi}^{\pi} \underbrace{g(x)}_{\text{even}} \underbrace{\sin nx}_{\text{odd}} dx \quad \text{odd} = 0$$

Since we are integrating an odd function over a symmetric interval.

Now Theorem 2.1 implies that for all x :

$$\checkmark \quad \underline{g(x) = a_0 + \sum_{n=1}^{\infty} (a_n \cos nx + b_n \sin nx)}$$
$$= \frac{1}{2}\pi + \sum_{n=1}^{\infty} \frac{2}{\pi} \left(\frac{1}{n^2} - \frac{(-1)^n}{n^2} \right) \cos nx$$

for example:



$$\int = -A + A = 0$$

$$\int_{-1}^0 (-x-1) dx$$
$$= -\frac{1}{2}x^2 - x \Big|_{-1}^0$$
$$= 0 - \left(-\frac{1}{2} + 1\right)$$
$$= 0 - \frac{1}{2}$$
$$= -\frac{1}{2}$$

2.1 Fourier Series: Examples

Solution (cont.)

$$g(x) = \frac{1}{2}\pi + \sum_{n=1}^N \frac{2}{\pi} \left(\frac{1}{n^2} - \frac{(-1)^n}{n^2} \right) \cos nx$$

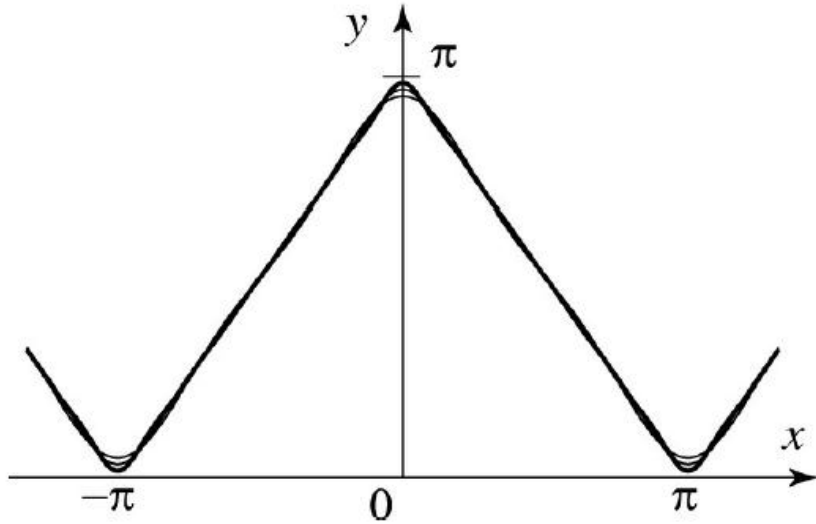


Figure 2.5 Partial sums of the Fourier series

$$g(x) = \frac{1}{2}\pi + \sum_{n=1}^{\infty} \frac{2}{\pi} \left(\frac{1}{n^2} - \frac{(-1)^n}{n^2} \right) \cos nx \quad (2.5)$$

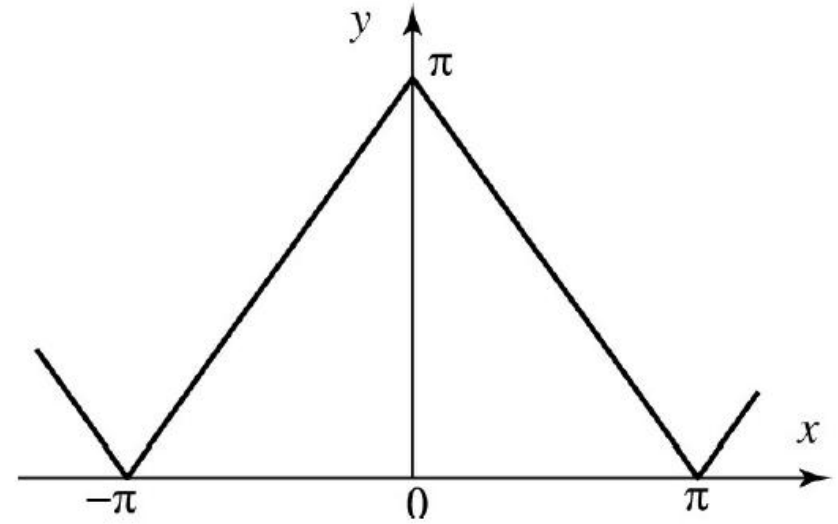


Figure 2.6 The Fourier series

Notice: Figure 2.5 is converging very fast, much faster than those in Example 1.3 of Lecture 1.

This is due to the magnitudes of the Fourier coefficients.

In Example 1.3 the coefficients are of the order $1/n$,

while in Example 2.2 the coefficients are of the order $1/n^2$.

2.1 Fourier Series: Examples

Operations on Fourier Series

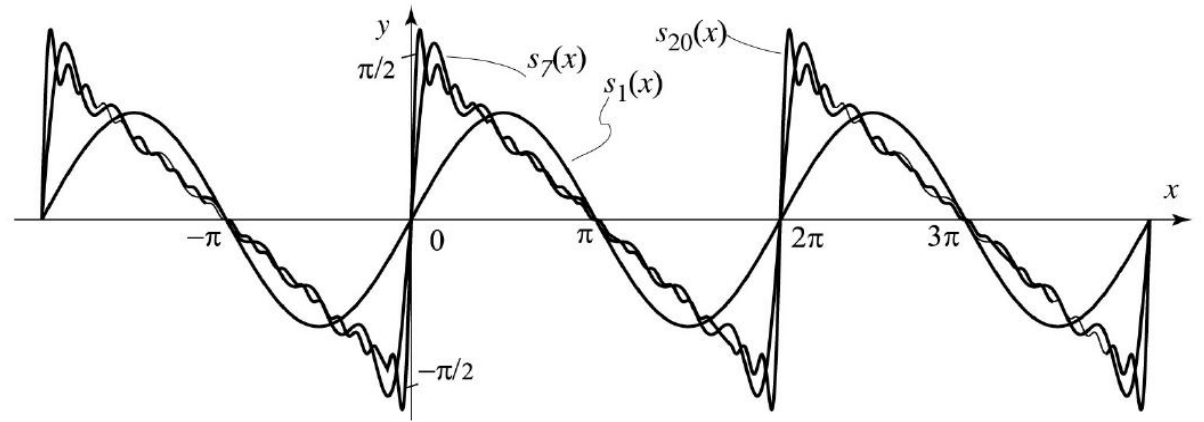


Figure 1.8 of Example 1.3

The Fourier series in [Examples 1.3](#) is an odd function and contains only sine terms, while the [Examples 2.2](#) is an even function and contains only cosine terms.

We will [derive the Fourier series containing both sine and cosine terms without computing Fourier coefficients but by applying operations such as multiplying \(乗算\) a Fourier series by a constant, adding \(加算\) two Fourier series, changing variables \(\$x\$ to \$-x\$ \), and translating \(平行移動\).](#)

2.1 Fourier Series: Examples

Operations on Fourier Series

$$\begin{aligned} g(x) &= \frac{1}{2}\pi + \sum_{n=1}^{\infty} \frac{2}{\pi} \left(\frac{1}{n^2} - \frac{(-1)^n}{n^2} \right) \cos nx \\ &= \frac{1}{2}\pi + \sum_{n \text{ odd}} \frac{4}{\pi n^2} \cos nx = \frac{1}{2}\pi + \frac{4}{\pi} \sum_{k=0}^{\infty} \frac{1}{(2k+1)^2} \cos (2k+1)x \end{aligned} \quad (2.5)$$

Example 2.3 Using Fourier series to sum series

If we take $x = 0$ on both sides of (2.5), we get

$$\pi = g(0) = \frac{1}{2}\pi + \frac{4}{\pi} \sum_{k=0}^{\infty} \frac{1}{(2k+1)^2}$$

From Figure 2.4

Subtracting $\frac{1}{2}\pi$ from both sides and then multiplying by $\frac{1}{4}\pi$ we get the interesting identity

$$\frac{\pi^2}{8} = 1 + \frac{1}{3^2} + \frac{1}{5^2} + \frac{1}{7^2} + \dots,$$

which can be used to approximate π^2 (and hence also π).

2.1 Fourier Series: Examples

Operations on Fourier Series

Example 2.4 Linear combinations of Fourier series

The 2π -periodic function

$$h(x) = \begin{cases} \pi - x & \text{if } 0 < x \leq \pi \\ 0 & \text{if } \pi < x < 2\pi \end{cases}$$

is related to the functions in [Examples 1.3](#) and [2.2](#) by

$$h(x) = f(x) + \frac{1}{2}g(x)$$

Find the Fourier series of $h(x)$

2.1 Fourier Series: Examples

Solution

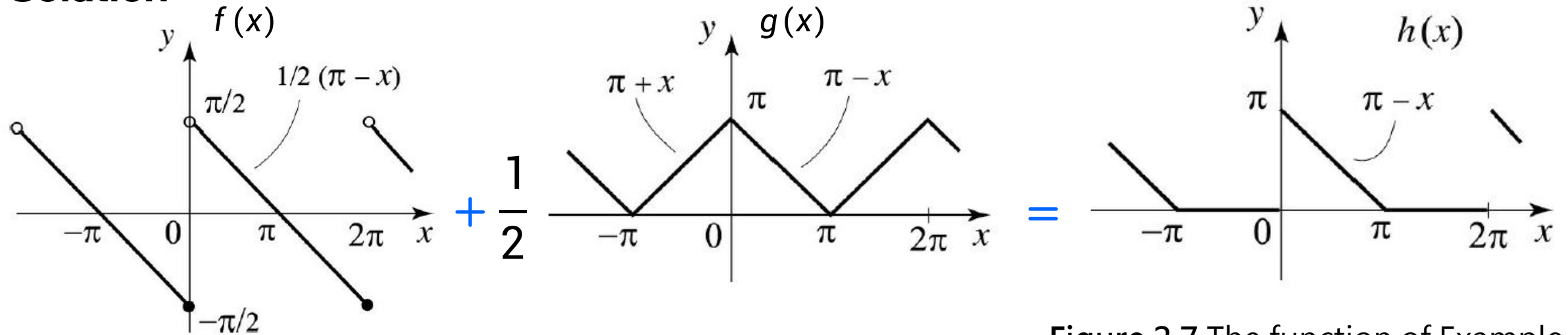


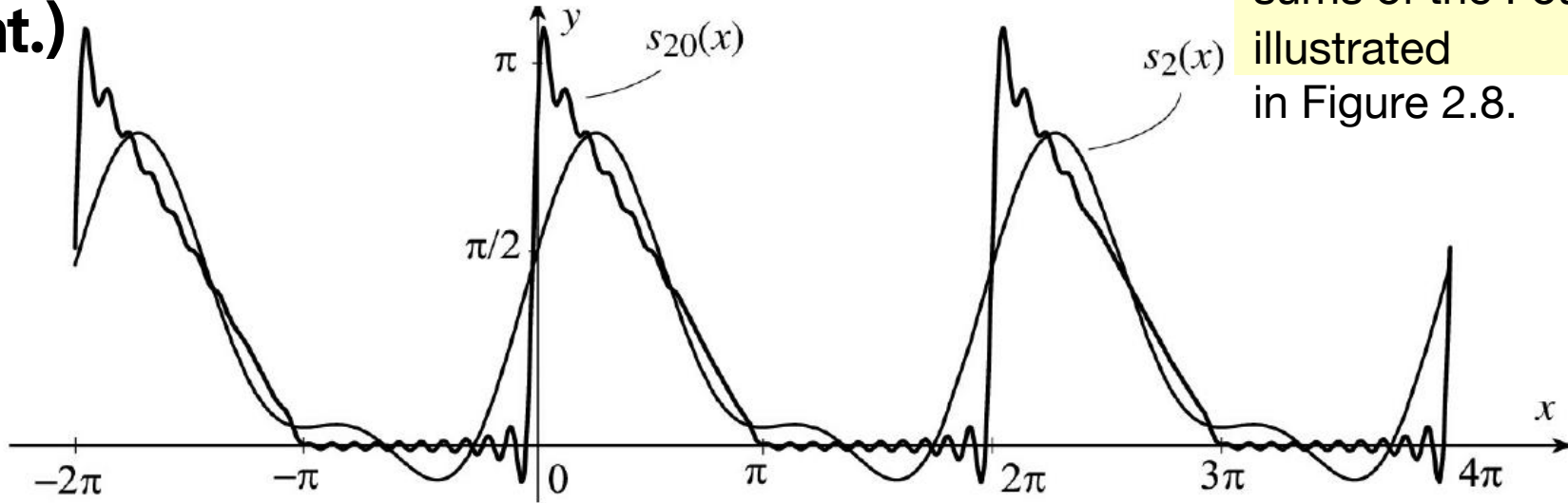
Figure 2.7 The function of Example 2.4

We can simply form the appropriate linear combination of the Fourier series.

$$\begin{aligned}
 h(x) &= f(x) + \frac{1}{2}g(x) \\
 &= \overbrace{\sum_{n=1}^{\infty} \frac{\sin nx}{n}}^{f(x)} + \overbrace{\frac{\pi}{4} + \frac{1}{\pi} \sum_{n=1}^{\infty} \left(\frac{1}{n^2} - \frac{(-1)^n}{n^2} \right) \cos nx}^{\frac{1}{2}g(x)}
 \end{aligned}$$

2.1 Fourier Series: Examples

Solution (cont.)



The **convergence** of the partial sums of the Fourier series to $h(x)$ is illustrated in Figure 2.8.

Figure 2.8 Note the Gibbs phenomenon **at the points of discontinuity** ($x = 2k\pi$, k is an integer). At these points, the Fourier series converges to the average value $\frac{\pi}{2}$. At all other points, the Fourier series converges to $h(x)$. This is due to the fact that the Fourier series consists of a cosine part that is converging very fast (Figure 2.5) and a sine part that overshoots at the points of discontinuity.

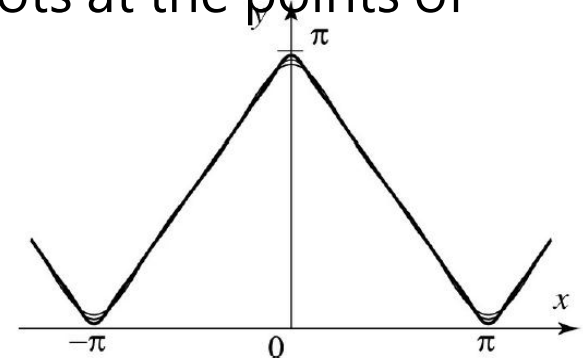


Figure 2.5 Partial sums of the Fourier series

2.1 Fourier Series: Examples

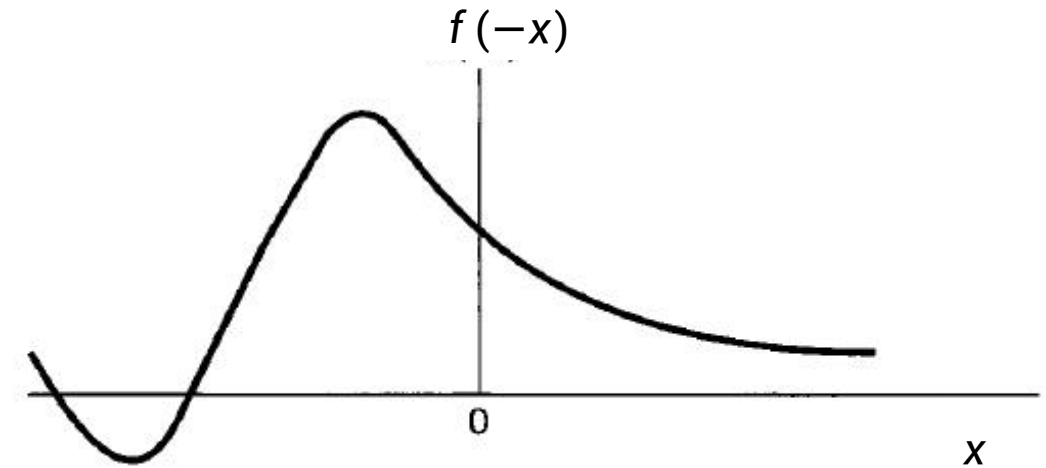
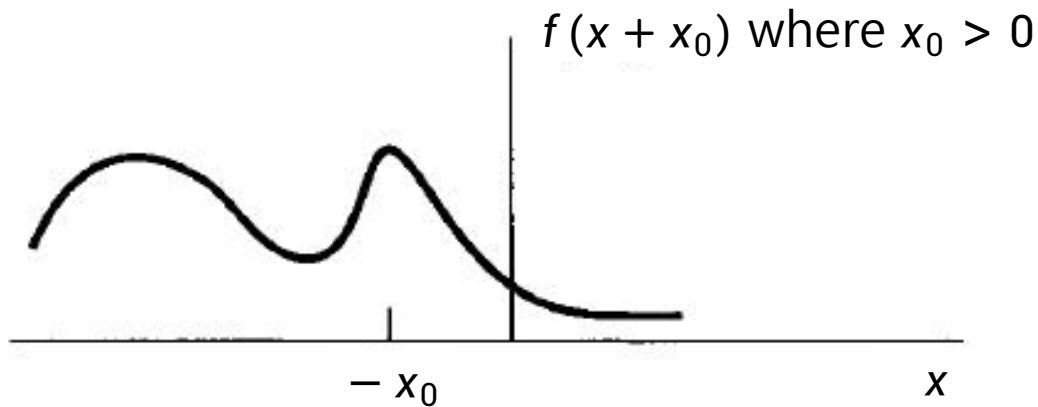
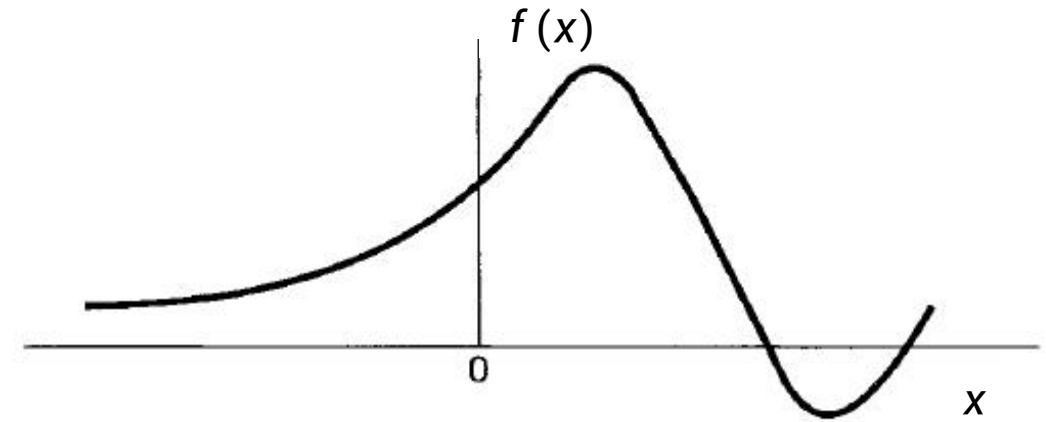
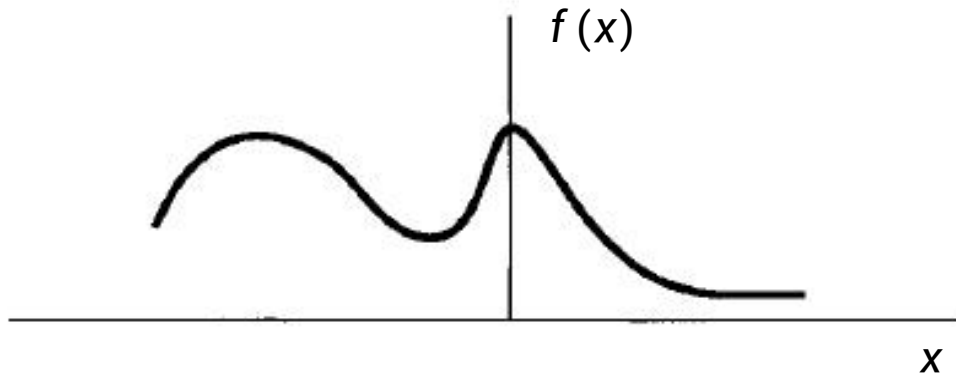


Figure 2.9 Translation (平行移動):

Two signals $f(x)$ and $f(x + x_0)$ that are identical in shape, but that are translated/shifted relative to each other

Figure 2.10 Reversal (逆転) (Changing Variable):

The signal $f(-x)$ is obtained from the signal $f(x)$ by a reflection about $x = 0$ (i.e., by reversing the signal).

2.1 Fourier Series: Examples

Exercise: Given the signal $f(x)$ shown in Figure 2.11.

Plot $f(x+1)$, $f(-x)$, $f(-x-1)$, and $f(-x+1)$.

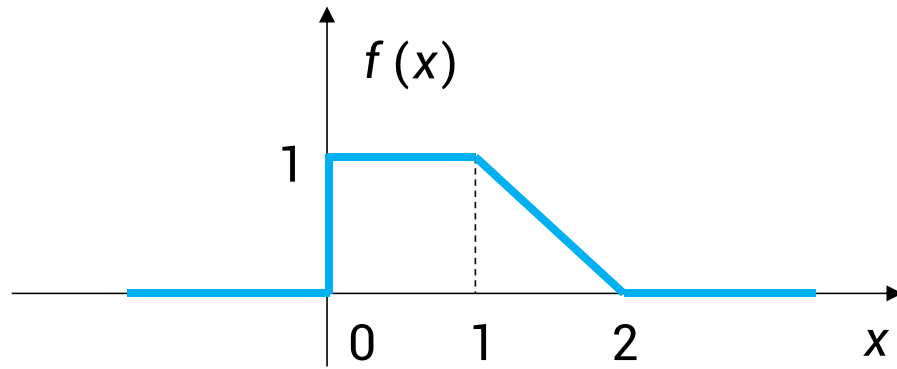
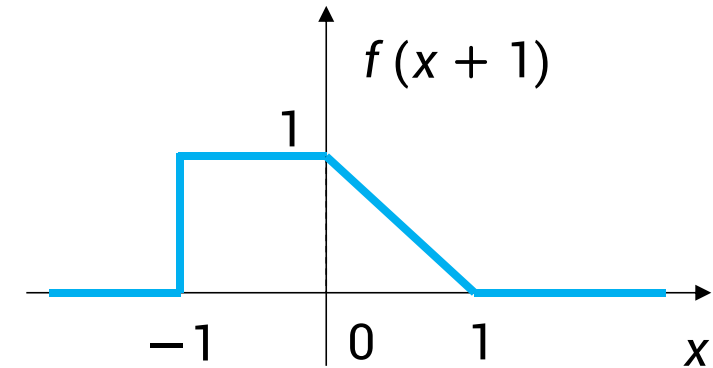
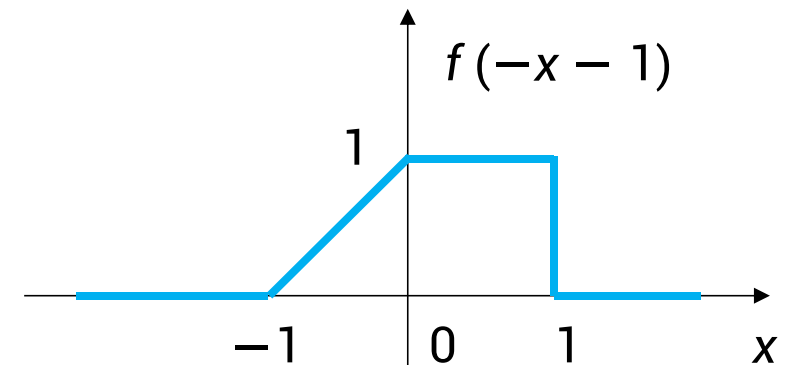
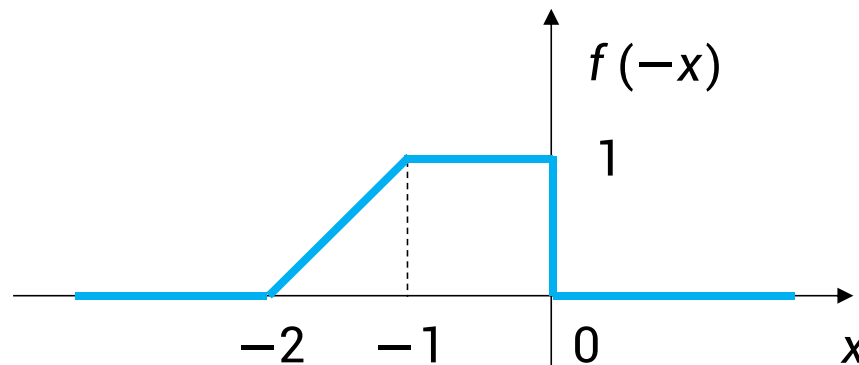
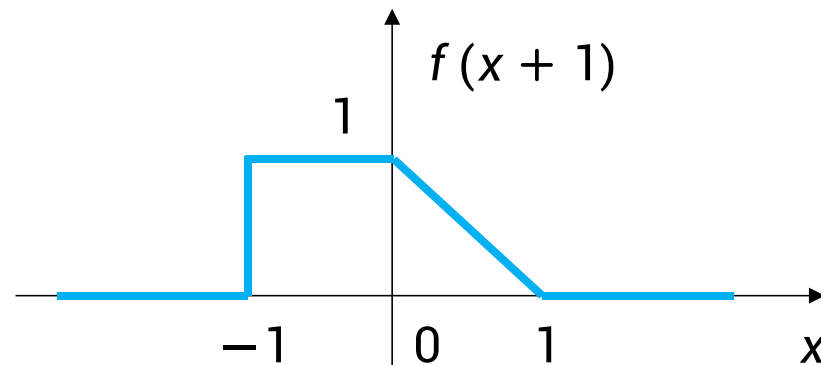


Figure 2.11

$f(-x-1)$
 $= f(-(x+1))$
Translate to left with
1 unit, then
reversing the signal.



Solution



2.1 Fourier Series: Examples

Exercise:

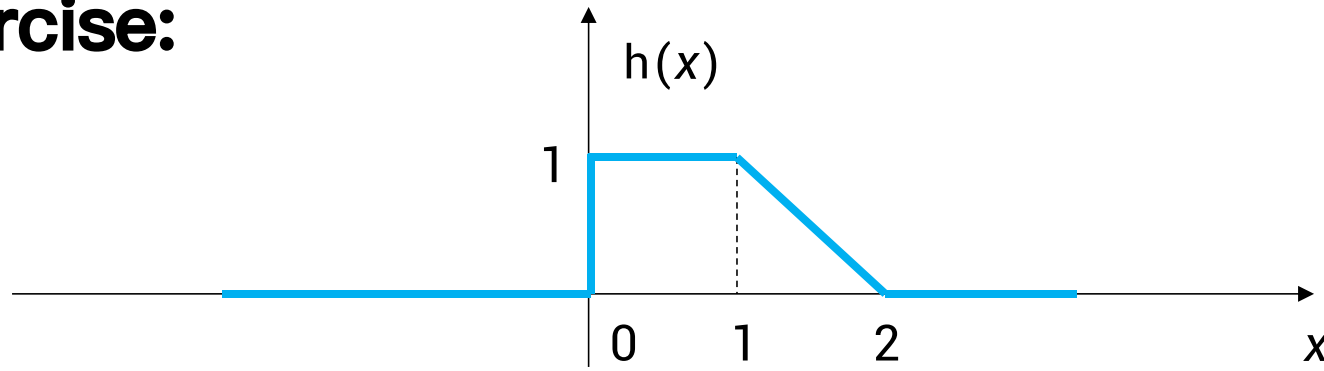


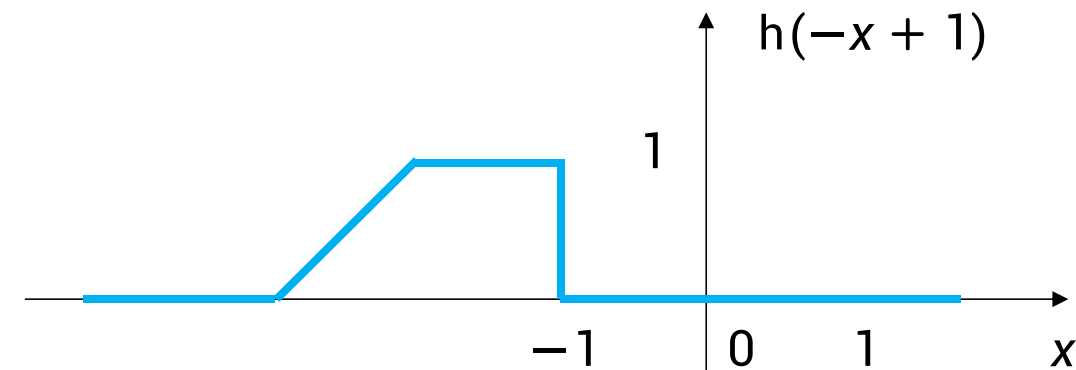
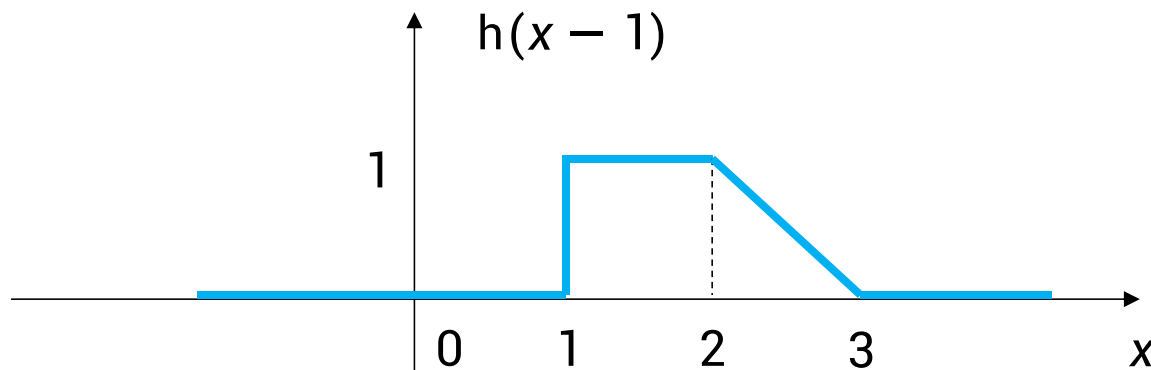
Figure 2.11

Solution

$$h(-x + 1)$$

$$= h(-(x - 1))$$

Translate to right
with 1 unit, then
reversing the signal.



2.1 Fourier Series: Examples

Example 2.5 Changing variables (変数の変更) and translating (平行移動)

The graph of the function $g(x)$ in Figure 2.12 is obtained by reflecting through the y -axis the graph in Figure 2.7 and then translating by π units to the left or right.

Find the Fourier Series of $g(x) = h(-x - \pi)$.

Solution: $g(x) = h(-x - \pi)$ and the Fourier series representation of $g(x)$ is

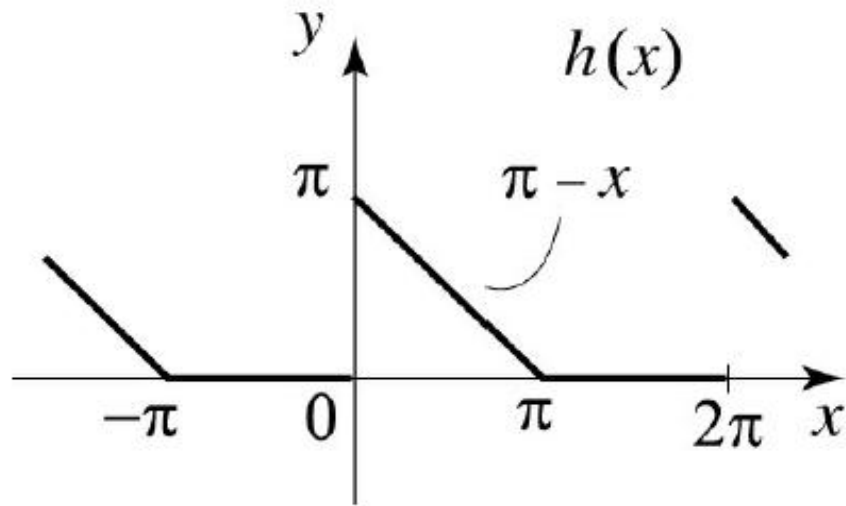


Figure 2.7 The function of Example 2.4

$$g(x) = h(-x - \pi)$$

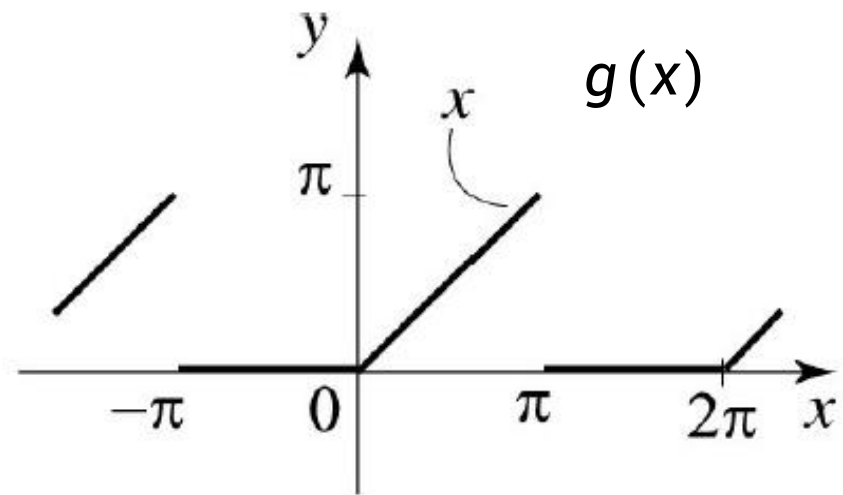



Figure 2.12 The function in Example 2.5.

2.1 Fourier Series: Examples

Example 2.5 Changing variables (変数の変更) and translating (平行移動)

The graph of the function $g(x)$ in Figure 2.12 is obtained by reflecting through the y -axis the graph in Figure 2.7 and then translating by π units to the left or right.

Find the Fourier Series of $g(x) = h(-x - \pi)$.

Solution (cont.):

The Fourier series representation of $g(x)$ is a transformations

from $h(x)$, therefore

$$g(x) = h(-x - \pi) = \frac{\pi}{4} + \sum_{n=1}^{\infty} \left[\frac{1}{\pi} \left(\frac{1}{n^2} - \frac{(-1)^n}{n^2} \right) \cos n(-x - \pi) + \frac{\sin n(-x - \pi)}{n} \right]$$

But $\cos n(-x - \pi) = (-1)^n \cos nx$ and $\sin n(-x - \pi) = (-1)^{n+1} \sin nx$. Thus

$$g(x) = \frac{\pi}{4} + \sum_{n=1}^{\infty} \left[\frac{1}{\pi} \left(\frac{(-1)^n}{n^2} - \frac{1}{n^2} \right) \cos nx + (-1)^{n+1} \frac{\sin nx}{n} \right]$$

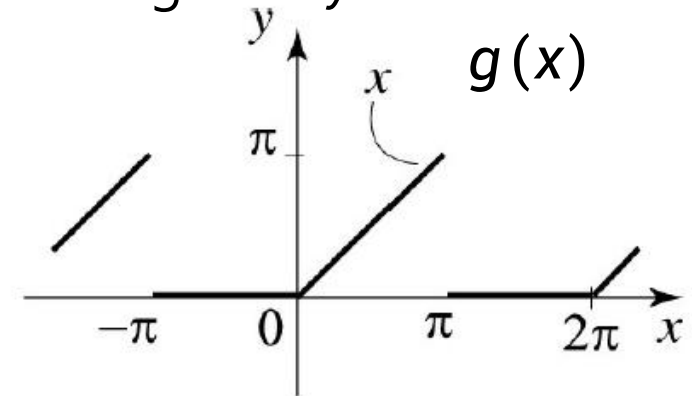


Figure 2.12

The function in Example 2.5.

2.2 Fourier Series of Functions with Arbitrary Periods

任意周期の関数のフーリエ級数

2.2 Fourier Series of Functions with Arbitrary Periods

In the preceding section we worked with functions of period 2π .

The choice of this period was merely for convenience.

We show how to extend our results to functions with arbitrary period (Figure 2.13) by using a simple change of variables.

Suppose that f is a function with period $T = 2p > 0$, and let

$$g(x) = f\left(\frac{p}{\pi}x\right) \quad (2.6)$$

Since f is $2p$ -periodic, we have

$$g(x + 2\pi) = f\left(\frac{p}{\pi}(x + 2\pi)\right) = f\left(\frac{p}{\pi}x + 2p\right) = f\left(\frac{p}{\pi}x\right) = g(x)$$

Hence g is 2π -periodic.

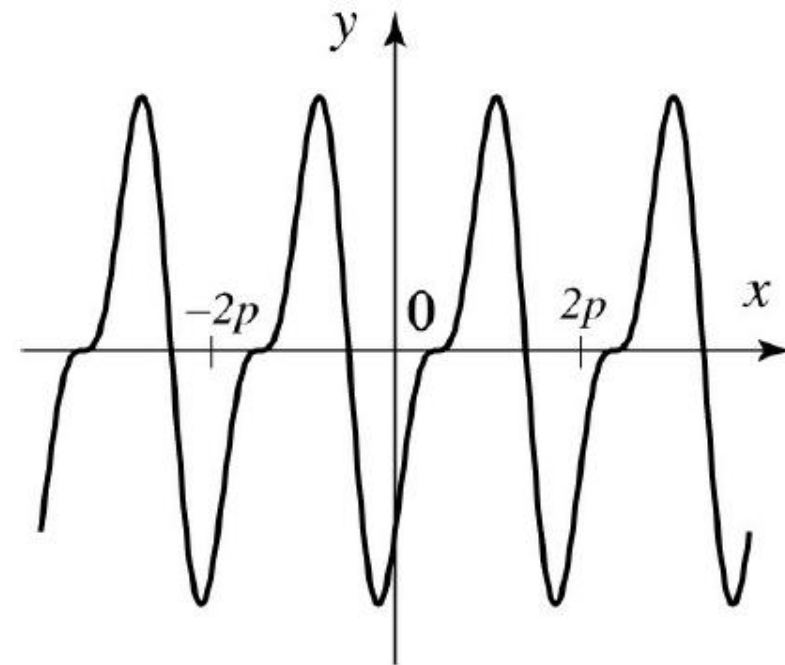


Figure 2.13 A $2p$ -periodic function

2.2 Fourier Series of Functions with Arbitrary Periods

Theorem 2.2 Fourier Series Representation: Arbitrary Period (任意周期)

Suppose that f is a **2p-periodic** piecewise smooth function. The Fourier series of f is given by

$$a_0 + \sum_{n=1}^{\infty} \left(a_n \cos \frac{n\pi}{p} x + b_n \sin \frac{n\pi}{p} x \right) \quad (2.7)$$

where

$$a_0 = \frac{1}{2p} \int_{-p}^p f(x) dx \quad (2.8)$$

$$a_n = \frac{1}{p} \int_{-p}^p f(x) \cos \frac{n\pi}{p} x dx \quad (n = 1, 2, \dots) \quad (2.9)$$

$$b_n = \frac{1}{p} \int_{-p}^p f(x) \sin \frac{n\pi}{p} x dx \quad (n = 1, 2, \dots) \quad (2.10)$$

The Fourier series converges to $f(x)$ if f is continuous at x and to $\frac{f(x-) + f(x+)}{2}$ otherwise.

By Theorem 1.1 of Lecture 1, all the integrals $\int_{-p}^p$ can be replaced by $\int_0^{2p}$ without changing the values of the coefficients.

Find the proof in the page 39 of textbook.

2.2 Fourier Series of Functions with Arbitrary Periods

Example 2.6 A Fourier series with arbitrary period

Find the Fourier series of the $2p$ -periodic function given by $f(x) = |x|$ if $-p \leq x \leq p$ ([Figure 2.14](#)).

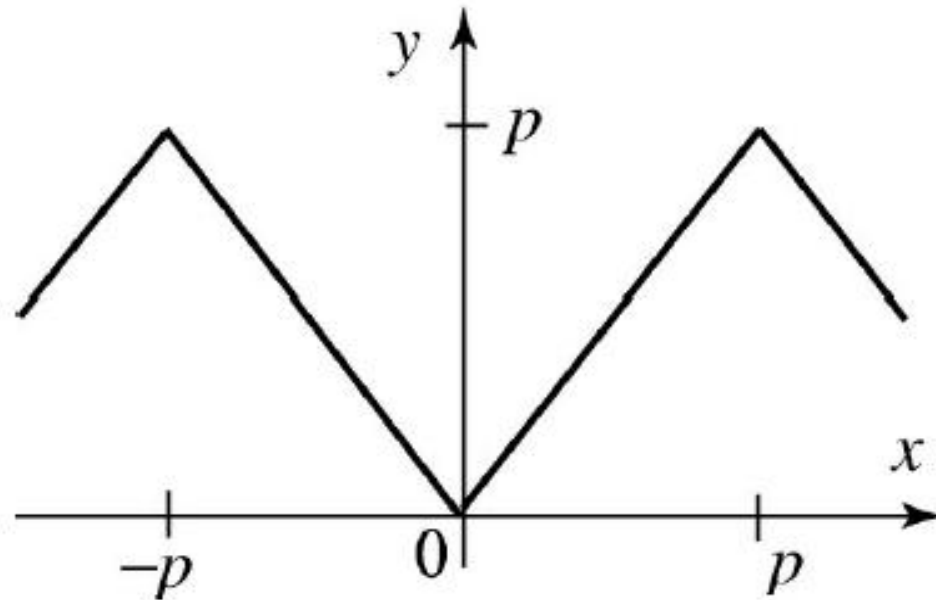


Figure 2.14 Triangular wave with period $2p$.

2.2 Fourier Series of Functions with Arbitrary Periods

Solution:

We compute the Fourier coefficient by using Theorem 2.2.

The area under the graph of f in Figure 2.14 gives.

$$a_0 = \frac{1}{2p} \int_{-p}^p f(x) dx = \frac{1}{2p} \left[\int_{-p}^0 -x dx + \int_0^p x dx \right] = \frac{1}{2p} \left[-\frac{1}{2}x^2 \Big|_{-p}^0 + \frac{1}{2}x^2 \Big|_0^p \right] \\ = \frac{1}{2p} \cdot p^2 = \frac{p}{2}$$

$$a_n = \frac{1}{p} \int_{-p}^p \overbrace{f(x)}^{\text{even function}} \cos \frac{n\pi}{p} x dx = \frac{1}{p} \cdot 2 \cdot \int_0^p f(x) \cos \frac{n\pi}{p} x dx \\ = \frac{2}{p} \int_0^p x \cos \frac{n\pi}{p} x dx \\ = \frac{-2p}{\pi^2 n^2} (1 - \cos n\pi) \quad (\text{integration by parts})$$

Since $\cos n\pi = (-1)^n$,

$$a_n = \begin{cases} 0 & \text{if } n \text{ is even} \\ -\frac{4p}{\pi^2 n^2} & \text{if } n \text{ is odd} \end{cases}$$

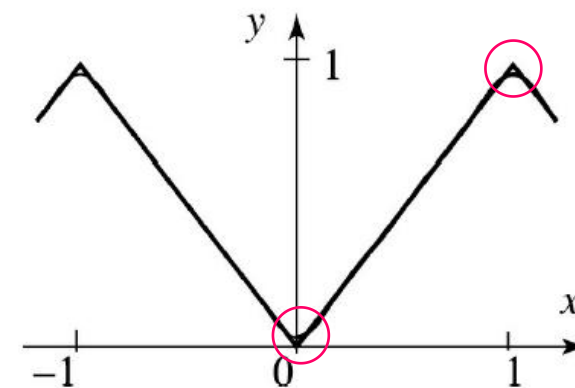


Figure 2.15 Partial sums of the Fourier series ($p = 1$), in Example 2.6.

2.2 Fourier Series of Functions with Arbitrary Periods

Since $\cos n\pi = (-1)^n$,

$$a_n = \begin{cases} 0 & \text{if } n \text{ is even} \\ -\frac{4p}{\pi^2 n^2} & \text{if } n \text{ is odd} \end{cases}$$

Similarly, since $f(x)$ is even, we have

$$b_n = 0$$

We thus obtain the Fourier Series

$$f(x) = \frac{p}{2} - \frac{4p}{\pi^2} \left(\cos \frac{\pi}{p} x + \frac{1}{3^2} \cos \frac{3\pi}{p} x + \frac{1}{5^2} \cos \frac{5\pi}{p} x + \dots \right) \quad f(x) = \frac{p}{2} - \frac{4p}{\pi^2} \sum_{k=0}^{\infty} \frac{1}{(2k+1)^2} \cos \left(\frac{(2k+1)\pi x}{p} \right)$$

Because f is continuous and piecewise smooth,

Theorem 2.2 tells us that Fourier series converges to $f(x)$

for all x . (See Figure 2.15)

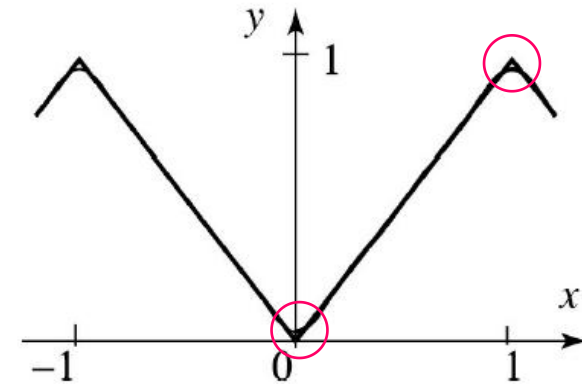


Figure 2.15 Partial sums of the Fourier series ($p = 1$), in Example 2.6.

2.2 Fourier Series of Functions with Arbitrary Periods

Even Function (偶関数) and Odd Function (奇関数)

Theorem 2.3 Fourier Series of Even and Odd Functions

Suppose that f is $2p$ -periodic and has the Fourier series representation (2.7). Then

(i) f is even if and only if $b_n = 0$ for all n . In this case

$$f(x) = a_0 + \sum_{n=1}^{\infty} a_n \cos \frac{n\pi}{p} x$$

where

$$a_0 = \frac{1}{p} \int_0^p f(x) dx \quad \text{and} \quad a_n = \frac{2}{p} \int_0^p f(x) \cos \frac{n\pi}{p} x dx \quad (n = 1, 2, \dots)$$

(ii) f is odd if and only if $a_n = 0$ for all n . In this case

$$f(x) = \sum_{n=1}^{\infty} b_n \sin \frac{n\pi}{p} x$$

where

$$b_n = \frac{2}{p} \int_0^p f(x) \sin \frac{n\pi}{p} x dx \quad (n = 1, 2, \dots)$$

2.2 Fourier Series of Functions with Arbitrary Periods

Even Function (偶関数) and Odd Function (奇関数)

Proof (i) If $f(x) = a_0 + \sum_{n=1}^{\infty} a_n \cos \frac{n\pi}{P} x$, then, for all x ,

$$f(-x) = a_0 + \sum_{n=1}^{\infty} a_n \cos \left(-\frac{n\pi}{P} x\right) = a_0 + \sum_{n=1}^{\infty} a_n \cos \frac{n\pi}{P} x = f(x)$$

and so f is even.

Conversely, suppose that f is even, use (2.4) and the fact that $\underbrace{f(x)}_{\text{even}} \underbrace{\sin \frac{n\pi}{P} x}_{\text{odd}}$ is odd to get that $b_n = 0$ for all n .

Use (2.8), (2.9), (2.3) and the fact that $f(x) \cos \frac{n\pi}{P} x$ is even to get the formulas for the coefficients a_0 and a_n in (i).

(ii) Similar to (i). (Omitted.)

2.2 Fourier Series of Functions with Arbitrary Periods

EXAMPLE 2.7 Fourier series of an **odd** function

Find the Fourier series of the **π -periodic function** $f(x) = x \cos x$ if $-\frac{\pi}{2} < x < \frac{\pi}{2}$.

Solutions:

From Theorem 2.3, its Fourier series is given by

$$f(x) = \sum_{n=1}^{\infty} b_n \sin \frac{n\pi}{p} x$$

where
$$b_n = \frac{2}{p} \int_0^p f(x) \sin \frac{n\pi}{p} x dx \quad (n=1, 2, \dots)$$

when $2p = \pi$, we have

$$f(x) = \sum_{n=1}^{\infty} b_n \sin 2nx$$

where
$$b_n = \frac{4}{\pi} \int_0^{\frac{\pi}{2}} x \cos x \sin 2nx dx$$

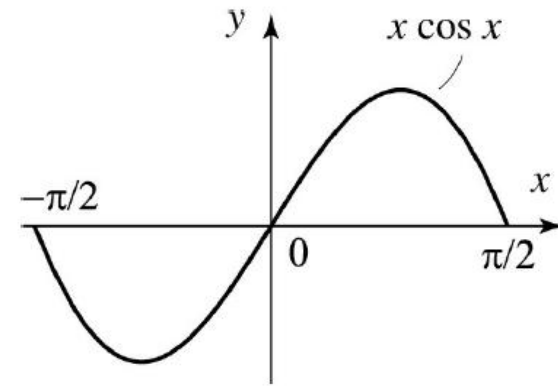


Figure 2.16 The odd function in Example 2.7.

To evaluate this integral, we will need the addition formula

$$\cos a \sin b = \frac{1}{2} [\sin(a+b) - \sin(a-b)]$$

and the integral formula

$$\int u \sin u du = \sin u - u \cos u + C$$

2.2 Fourier Series of Functions with Arbitrary Periods

EXAMPLE 2.7 Fourier series of an **odd** function

Find the Fourier series of the **π -periodic function** $f(x) = x \cos x$ if $-\frac{\pi}{2} < x < \frac{\pi}{2}$.

Then we have

$$\begin{aligned}
 b_n &= \frac{4}{\pi} \int_0^{\frac{\pi}{2}} x \left\{ \frac{1}{2} [\sin(1+2n)x - \sin(1-2n)x] \right\} dx \\
 &= \frac{2}{\pi} \left\{ \int_0^{\frac{\pi}{2}} x \sin(2n+1)x dx + \int_0^{\frac{\pi}{2}} x \sin(2n-1)x dx \right\} \\
 &= \frac{2}{\pi} \cdot \frac{1}{(2n+1)^2} \left[\sin(2n+1)x - (2n+1)x \cos(2n+1)x \right]_0^{\frac{\pi}{2}} \\
 &\quad + \frac{2}{\pi} \cdot \frac{1}{(2n-1)^2} \left[\sin(2n-1)x - (2n-1)x \cos(2n-1)x \right]_0^{\frac{\pi}{2}} \quad (\text{integration by parts}) \\
 &= \frac{2}{\pi(2n+1)^2} \sin(2n+1)\frac{\pi}{2} + \frac{2}{\pi(2n-1)^2} \sin(2n-1)\frac{\pi}{2} \\
 &\quad \text{Since } \sin(2n+1)\frac{\pi}{2} = (-1)^n \text{ and } \sin(2n-1)\frac{\pi}{2} = (-1)^{n+1} \\
 &= \frac{2}{\pi} (-1)^n \left[\frac{1}{(2n+1)^2} - \frac{1}{(2n-1)^2} \right] \\
 &= \frac{16}{\pi} (-1)^{n+1} \frac{n}{(2n+1)^2 (2n-1)^2}
 \end{aligned}$$

Thus

$$f(x) \approx \frac{16}{\pi} \left[\frac{1}{9} \sin 2x - \frac{2}{225} \sin 4x + \dots \right]$$

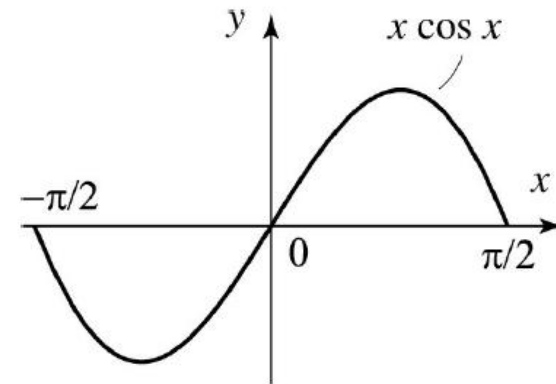


Figure 2.16 The odd function in Example 2.7.

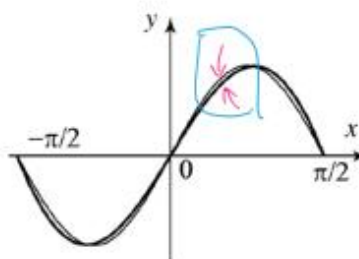


Figure 2.17 Graphs of $f(x)$, $s_2(x)$ and $s_4(x)$ in Example 2.7.

Review for Lecture 2

- Gibbs Phenomenon
- Even Function and Odd Function
- Operations on Fourier Series
- Fourier Series of Functions with Arbitrary Periods

Assignment

Please Check <https://web-ext.u-aizu.ac.jp/~xiangli/teaching/MA05/index.html>

Reading materials: Lecture 2 Slides

References

- [1] Nakhlé H. Asmar, *Partial Differential Equations with Fourier Series and Boundary Value Problems 2nd Edition*, 2004
- [2] Wikipedia